

- **Pilot light:** Maintain a read-only replica of the primary database in the recovery region. The replica is kept up-to-date using replication technology, and it can be quickly promoted to a read-write instance if the primary database becomes unavailable.
- **Warm standby:** Maintain a warm standby database in the recovery region. The standby database is kept in a near-ready state, with data refreshed periodically or on-demand. It can be activated as the primary database with minimal downtime in case of a disaster.
- **Multi-site active/active:** Operate active-active production databases in multiple AWS regions. Data is replicated between the regions in real-time, allowing failover to either region with minimal disruption.

The choice of DR strategy depends on factors such as recovery time objective (RTO), recovery point objective (RPO), cost, and complexity.

RTO: This is the time acceptable for downtime following a disaster. Backup and restore strategies may have longer RTOs compared to pilot light or warm standby.

RPO: This is the acceptable amount of data loss following a disaster. Warm standby and active/active strategies minimise data loss compared to backup and restore.

Cost: Backup and restore is the most cost-effective DR strategy, while active/active is the most expensive. Pilot light and warm standby offer a balance between cost and performance.

Complexity: Backup and restore is the simplest, while active/active is the most complex to implement and manage. **END** 🐧

References

- <https://aws.amazon.com/memorydb/>
- <https://aws.amazon.com/big-data/featured-partner-solutions-data-governance-compliance/>
- <https://aws.amazon.com/data-exchange/providers/>
- <https://aws.amazon.com/aws-cost-management/aws-cost-optimization/right-sizing/>
- <https://aws.amazon.com/datapipeline/>
- <https://aws.amazon.com/athena/>
- <https://aws.amazon.com/blogs/architecture/disaster-recovery-dr-architecture-on-aws-part-i-strategies-for-recovery-in-the-cloud/>

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Continued from page...37

Advances in flash storage: The continued development of flash storage technology is likely to impact SAN and NAS design, as flash storage can offer significantly faster data access speeds compared to traditional storage solutions.

Overall, forecasting the future of SAN and NAS design involves considering factors such as data growth, emerging technologies, and shifting business needs to predict how storage solutions will need to evolve to remain effective.

Final thoughts

Here are some final thoughts and recommendations for open source

SAN and NAS design.

- Open source SAN and NAS solutions provide businesses with cost-effective and flexible storage options. They offer a significant cost advantage over proprietary solutions and can be customised to meet unique business needs.
- These solutions are fully transparent, giving users complete control over configurations, updates, and upgrades.

- Community support is essential for open source SAN and NAS solutions. A strong community can help refine the design and functionality of these solutions, which ultimately benefits everyone.
- Security is of utmost importance in SAN and NAS design. User and data safety must be maintained, and open source solutions must keep pace with existing security threats. **END** 🐧

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